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15 December 1986
SUPERSEDING
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MILITARY SPECIFICATION

CANS, WATER, MILITARY, 5-GALLON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers two types of rectangular 5-gallon water cans equipped with carrying handles.

1.2 Classification. Water cans shall be of the following types, as specified (see 6.2).

Type I - Steel
Type II - Aluminum

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7240

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SPECIFICATIONS

FEDERAL

A-A-665	-	Lacquer (Aerosol)
HH-C-576	-	Cork Sheet (Gaskets, Sheets, and Strips)
QQ-T-371	-	Tin, Pig
TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings
TT-E-485	-	Enamel, Semi-Gloss, Rust-Inhibiting
TT-E-529	-	Enamel, Alkyd, Semi-Gloss
TT-P-1757	-	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity
PPP-F-320	-	Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

MILITARY

MIL-C-5541	-	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-L-52043	-	Lacquer, Semi-Gloss, Cellulose Nitrate

STANDARDS

FEDERAL

FED-STD-595	-	Colors

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads

DRAWINGS

US ARMY NATICK RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

3-3-22	-	Can, Water, Military, 5-Gallon, Steel; Assembly
3-3-23	-	Section Details
3-3-24	-	Cover Assembly
3-3-25	-	Closure Assembly
3-3-57	-	Can, Water, Military, 5-Gallons, Aluminum Assembly
3-3-58	-	Cover Assembly
3-3-59	-	Sections
5-11-809	-	Can, Water, Military, 5-Gallon; Strapping for Shipment

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(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal Agency responsible for the issuance thereof.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issues of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 366	-	Steel, Carbon, Cold Rolled Sheet, Commercial Quality
A 569	-	Steel, Carbon, Hot-Rolled Sheet and Strip, Commercial Quality
B 633	-	Electrodeposited Coatings of Zinc on Iron and Steel
D 2000	-	Rubber Products in Automotive Applications

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

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3.2 Materials and components. Materials and components specified shall conform to applicable documents specified herein. Materials and components not definitely specified shall be of the quality currently used by the contractor provided the completed item complies with all provisions of this document (see 6.4).

3.3 Toxic products and formulations. The interior of the finished water cans shall contain no toxic materials or materials containing toxic components. Prior to use, the contractor shall submit to the contracting officer or his authorized representative certification that the tin flux, seam cement, cork binder, gasket cement, and interior coating are non-toxic and will have no adverse effect on the health of personnel when used for the intended purposes. The certified statement shall include information as to formulation in sufficient detail to establish nontoxicity. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate department medical service who will act as an advisor to the contracting activity.

3.3.1 Tin, pig. Pig tin shall conform to QQ-T-371.

3.3.2 Cork. The gasket for type II can shall be fabricated from cork-compositions conforming to HH-C-576, except that the binder used shall be nontoxic (see 3.3) for use in the packaging of food products.

3.3.3 Cement, seam. Seam cement shall be a nontoxic (see 3.3) commercial grade of the nonhardening type and shall withstand the baking process required after applying the coatings specified in 3.5.3.4.

3.3.4 Primer. Primer shall conform to composition G or L, as applicable, color Y of TT-P-1757.

3.3.5 Exterior coating

3.3.5.1 Can type I, exterior coating. Exterior coating for the type I can shall be enamel conforming to TT-E-485 or a polyester powder coating. The color shall be Olive drab, color number 24087 of FED-STD-595.

3.3.5.2 Can type II, exterior enamel. Exterior enamel for the type II can shall conform to TT-E-529. The color shall be Olive drab, color number 24087 of FED-STD-595.

3.3.6 Can type I, interior coating. Interior coating for the type I can shall be gloss enamel or epoxy powder coating of a light cream, light tan, or ivory color and shall be a flexible, non-toxic (see 3.3) baked type coating based on phenolic or epoxy resins. Ingredients of the coating shall comply with Part 175.300 - Resinous and Polymeric Coatings of Chapter I of 21 CFR - Food and Drugs. When tested for resistance of acid, chlorine, trisodium phosphate, steam, and freeze-thaw, as specified in 4.4.2.4.1, the coating shall show no blistering, lifting, softening, wrinkling, checking, cracking, or signs of rusting through the coating. The coating, when applied to can interiors as specified in 3.5.3.4.1.1, shall meet the performance requirements specified therein.

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3.3.7 Lacquer. Lacquer shall conform to the requirements of A-A-665, or composition G or L of MIL-L-52043. The color shall be black.

3.3.8 Rubber. Rubber for the gasket of the type I can shall meet the requirements of type, class, and suffix letters 4AA520A13B13Z1Z2 of ASTM D 2000. Requirements Z1 and Z2 indicate special requirements. The requirement for Z1 is that rubber shall be certified to be non-toxic in accordance with Code of Federal Regulations, Title 21 - Food and Drugs, Chapter I - Food and Drug Administration, Department of Health, Education and Welfare, Part 177 - Indirect Food Additives: Polymers: Sub Part C - Substances For Use Only As Components of Articles Intended for Repeated Use; Section 177.2600 - Rubber Articles Intended for Repeated Use. The Z2 requirement is that the rubber shall be light cream, light tan, or ivory color.

3.4 Design and construction. The design and construction of the water cans shall be as specified herein, and as shown on the drawings specified in 2.1.

3.4.1 Can, type I.

3.4.1.1 Seams. All seams shall not leak when tested under air pressure as specified in 4.4.2.4.2. Leaks shall be repaired by the same method used in fabricating the seam assembly. The vertical lap seam shall be welded completely along and across the area of overlap. The vertical seam shall be on the end of the can opposite the pouring spout. The double seam, joining the bottom to the body, shall be evenly and tightly clinched. The double seam shall be sealed with the seam cement specified in 3.3.3. There shall be no excess cement remains on the inside or outside of the can. The seam joining the top to the body shall be lap welded with a continuous weld. The width of the finished seam shall be not less than 0.25 inch. The seam shall be welded completely along and across the area of overlap. The handle spot welds shall not pull away from the can when tested as specified in 4.6.12.

3.4.1.2 Tinning and assembly of neck to top. Prior to assembly of the neck to the top, the inside edge of the flanged opening shall be free of cracks and shall have all burrs and sharp edges removed. The neck assembly shall be sweated to the flanged opening in the top of the can by use of the pig tin specified in 3.3.1. The outside surface of the flange in the top and the inside surface of the neck, extending over the top of and approximately half way down the outside of the loop holding the lock ring, shall be tinned to a thickness not less than 0.002 inch. The tin shall be tightly adherent to the base metal, shall completely cover the above specified surfaces, and shall be free from bare spots, scratches, burrs, and abrasions.

3.4.2 Can type II, seams. The interior and exterior edges of the vertical seam, after lap welding, shall be without spaces or voids and the weld shall be complete along and across area of overlap. The vertical seam shall be on the edge opposite the pouring spout. The seams joining the top and bottom to the body shall be welded with a continuous weld and the width of weld shall be

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not less than 0.25 inch. The neck assembly shall be joined to the top with a double lap seam. All seams shall not leak when tested under air pressure as specified in 4.4.2.4.2. The handle spot welds shall not pull away from the can when tested as specified in 4.6.12.

3.4.3 Cover assemblies. The cover assemblies (closures) shall be constructed and assembled so that when the locking lever cams engage the cams on the lock rings and are pressed into closed position, sufficient pressure is applied to the gasket to ensure a watertight seal. The cover assemblies shall show no visible evidence of water leaks when tested as specified in 4.4.5.

3.4.4 Gaskets. The gaskets for the type I can shall be fabricated of rubber specified in 3.3.8. The gaskets for the type II can shall be fabricated of cork specified in 3.3.2. The gaskets shall be secured to the inside of the can cover with a nontoxic cement after application of the finished coats.

3.5 Finish.

3.5.1 Preliminary preparation. All metal splinters, welding flashes, burrs, and rough or sharp edges on the interior and exterior surfaces of the cans shall be removed by grinding, filing, or other effective methods to render the surface smooth in preparation for finishing.

3.5.2 Can type I, plating. The locking lever cam stud, hinge pin, and washers shall be zinc plated prior to assembly. Zinc plating shall conform to type IV (with a minimum thickness of 13 micrometer) of ASTM B 633.

3.5.2.1 Can type II, plating. The locking level cam stud, hinge pin, and washer shall be given a commercial hot-dip zinc coat. The neck sub-assembly (lock ring and hinge bracket), shall be electro-zinc coated after welding, and before assembly to can neck. Coating shall conform to type IV (with a minimum thickness of 13 micrometer) of ASTM B 633.

3.5.3 Surface preparation.

3.5.3.1 Can type I. All flux shall be removed and all interior and exterior surfaces of type I cans shall be given a prepaint treatment conforming to type I of II of TT-C-490.

3.5.3.2 Can type II.

3.5.3.2.1 Exterior surfaces. Prior to painting, all exterior aluminum surfaces of type II cans shall be prepared for an organic finish by application of class 1A film of MIL-C-5541. The cover of the can shall be prepared by this method but it shall be accomplished prior to assembly of the cover to the hinge bracket.

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3.5.3.2.2 Interior surfaces. After exterior surface preparation specified in 3.5.3.2.1, the interior of type II cans shall be thoroughly cleaned with a 10-percent solution of sulfuric acid. The cans shall then be thoroughly flushed with 150° to 200°F temperature water followed by a 3-minute flushing with live steam and then dried. The water in the cleaned cans shall be neutral in taste when tested as specified in 4.4.5.

3.5.3.3 Can type II, primer coating. Exterior aluminum surfaces of the type II can, including cover after cleaning and film application specified in 3.5.3.2.1, shall be coated with primer specified in 3.3.4. Before assembly to the can, the neck sub-assembly (lock ring, cam locking lever, hinge pin, washers, and studs) shall also be coated with primer specified in 3.3.4.

3.5.3.4 Coating.

3.5.3.4.1 Interior.

3.5.3.4.1.1 Can type I, coating interior. The vertical and horizontal welded seams in the interior of type I cans shall be stripe-coated with coating specified in 3.3.6 (if epoxy powder coating is used stripe-coating is not required). The stripe-coating shall be done by any method which ensures full coverage of the edge and surrounding area of overlapped metal sheet. Edges of seams shall show no voids when examined with a drumlight and mirror. The entire interior, including the inner face of the closure, shall then be coated with coating specified in 3.3.6. Special care shall be taken to ensure that the countersink of the can bottom and the coating shall be applied and baked in accordance with the coating supplier's instructions. The coating shall be baked prior to double seaming the can bottom. When tested after double seaming and all subsequent operations, the interior coating of the complete can shall meet the following requirements:

a. Impact flexibility. When tested as specified in 4.4.5, there shall be no cracking, checking, flaking, or loosening of the coating.

b. Resistance to hot water. When tested as specified in 4.4.5, there shall be no softening, blistering in excess of that defined in 4.6.10, separation, loosening, or other visible change in the coating and no evidence of rust deposited on, or penetrating through, the coating.

c. Taste. The water shall be neutral in taste when tested as specified in 4.4.5.

3.5.3.4.2 Exterior.

3.5.3.4.2.1 Can type I, coating exterior. After the application and baking of the interior finish of type I can, the exterior surfaces of the can, including the closure, shall be thoroughly coated with the coating specified in 3.3.5.1. The coating shall level out to a smooth and uniform dry film. The coating thickness when tested as specified in 4.4.5 shall be not less than 1 mil.

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3.5.3.4.2.2 Can type II, enamel coating exterior. After priming as specified in 3.5.3.3, the exterior surfaces of can and cover, and the surfaces of the neck sub-assembly shall be thoroughly coated with enamel specified in 3.3.5.2. Enameling of the neck sub-assembly, locking lever cam, and cover shall be applied prior to assembly to the can. Additional enameling of these parts after assembly is permissible. The combined film thickness of the primer and enamel when tested as specified in 4.4.5 shall be 1.4 mil minimum.

3.6 Marking. Each can shall be marked with the word "WATER" using black lacquer specified in 3.3.7. Each can shall be permanently indented with the manufacturer's name, trade mark, and the number and year of contract in characters not less than 0.25 inch high.

3.7 Workmanship. All components and assemblies of the water cans shall be free from dirt and other harmful extraneous materials, burrs, slivers, rough die marks, tool and grind marks, dents, and cracks. External surfaces shall be free from sharp edges and corners.

3.7.1 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be free of cracks, burn holes, and incomplete fusion. All scale and flux shall be removed from the finished weld area.

3.7.2 Rivets. Rivets shall be driven to completely fill the holes, shall have full size beads neatly finished and in full contact with the surface of the members, and shall not be loose.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1), it shall be examined for the defects listed in 4.4.3 and 4.4.4 and tested as specified in 4.6.2, 4.6.3, 4.6.9, 4.6.10, 4.6.11 and 4.6.12. The presence of any defect or the failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 In-process inspection.

4.4.2.1 In-process inspection . Inspection shall be made of the operations and requirements listed in table I to establish conformance with specified requirements. Whenever nonconformance is noted, correction shall be made to the items affected and to the operation.

TABLE I. In-Process inspection

Requirement or operation	Requirement paragraph
Vertical lap seam welded completely along and across area of overlap (type I)	3.4.1.1
Bottom seam cement application (type I) as specified	3.4.1.1

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TABLE I. In-Process inspection (cont'd)

Requirement or operation	Requirement paragraph
Upper horizontal lap seam welded completely along and across area of overlap (type I) as applicable	3.4.1.1
Burrs and sharp edges removed from flanged opening	3.4.1.2
Edge of flanged area free of cracks	3.4.1.2
Neck assembly sweated to can top (type I) as specified	3.4.1.2
Lap welded seam edges (interior and exterior) are free of spaces or voids (type II)	3.4.2
Gasket assembled to cover with cement as specified	3.4.4
Free from metal splinters, welding flash, burrs, rough and sharp edge (interior and exterior)	3.5.1
Plating of steel component parts (type I and II) as specified	3.5.2 and 3.5.2.1
Cleaning and pretreatment (type I)	3.5.3.1
Film application on exterior can and cover surfaces (type II)	3.5.3.2.1
Interior surface cleaning (type II)	3.5.3.2.2
Interior surface coating baked prior to double seaming (type I)	3.5.3.4.1.1
Interior surface coating baked according to paint suppliers recommendation (type I).	3.5.3.4.1.1
Seam striping (type I, not applicable to epoxy powder coating.)	3.5.3.4.1.1
Drum light examination (type I).	3.5.3.4.1.1
Welds free from rust scale, paint, grease, foreign matter, cracks, burn holes or incomplete fusion, scale and flux removal.	3.7.1

4.4.2.2 In-process primer inspection of type II cans and component parts. The type II cans and the component parts, prior to assembly to the can, shall be examined for the defects listed in table II. The lot shall be all primed cans, and all neck sub-assemblies, lock ring and hinge brackets, cam locking

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levers, covers, hinge pins, washers, and studs offered for inspection at one time. The sample unit shall be one each of the can, neck sub-assembly, cam locking lever, cover, hinge pin, washer, and stud. The inspection level (applicable to cans and component parts separately) shall be S-1 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 6.5.

TABLE II. Primer application defects

Examine	Defect
Primer application (type II cans and components)	Coating not uniform, continuous, or evenly deposited Not free of stains or smut Not free of blisters, sags, runs Not primed before assembly Primer on interior surface of cover or gasket

4.4.2.3 In-process dimensional examination. In-process dimensional examination shall be made of type I and II cans for: wall thickness (before finishing), lap welded seams, neck sub-assembly, cam locking levers, covers, hinge pins, washers, and studs. Any dimension that is not within the specified tolerances shall be classified a defect. The lot shall be all cans, and all neck sub-assemblies, cam locking levers, covers, hinge pins, washers, and studs offered for inspection at one time. The sample unit shall be one each of the can, neck sub-assembly, cam locking lever, cover, hinge pin, washer, and stud. The inspection level (applicable to cans and component parts separately) shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.2.4 In-process testing.

4.4.2.4.1 Steel panel coating (type I cans only). Prior to commencing production, five steel panels which have had both sides prepared as specified in 3.5.3.1 and 3.5.3.4.1.1 shall be tested for the characteristics listed in table III. Each panel shall be 4 by 6 inches, 20 gauge hot rolled or cold drawn quality sheet steel conforming to ASTM A 366 or A 569. Dry film thickness of the coating shall be not less than 0.8 mils. The coated and cured panels shall be free of fatty edges, sages, or other application defects and shall be aged 24 hours at 73.5 $\pm$ 4°F and 50 $\pm$ 5 percent relative humidity before testing. A separate panel shall be used for each test. Any nonconformance with specified requirements shall be cause for correction of the coating process.

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TABLE III. Steel panel in-process tests

Characteristic	Requirement paragraph	Test method
Resistance to acid	3.3.6	4.6.4
Resistance to chlorine	3.3.6	4.6.5
Resistance to tri-sodium phosphate	3.3.6	4.6.6
Resistance to steam	3.3.6	4.6.7
Resistance to freeze thaw	3.3.6	4.6.8

4.4.2.4.2 Air pressure leakage (type I and type II). Every can shall be tested for air pressure leakage as specified in 4.6.1, after assembly but prior to finishing. Any can failing to conform to 3.4.1.1 (type I) or 3.4.2 (type II), as applicable, shall be rejected.

4.4.2.4.3 Handle spot welds for types I and II. The can top handle assembly shall be tested for weld adhesion by testing as specified in 4.6.12. The inspection level shall be S-2 and any test failure shall be cause for rejection of all handle can top assemblies processed to the time of inspection and shall be cause for correction of the welding operation/process.

4.4.3 End item visual examination. The end item shall be examined for the defects listed in table IV. The lot size shall be expressed in units of cans. The sample unit shall be one can. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE IV. End item visual defects

Examine	Defect	Classification Major	Classification Minor
Finish, exterior, type I and II	Rust or corrosion (steel parts)		X
	Wrong color, shade		X
	Dirt, grit, or foreign matter imbedded in finish		X
	Oil, flux, or seam cement showing through finish (exterior)		X
	Enamel not dry	X	
	Enamel blistered, peeling, not adhering		X
	Soft run, sag, or tear drop		X

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Finish, interior, type I and II (examine with drum light and mirror)	Rust or corrosion	X	
	Interior of neck not free of exterior enamel	X	
	Exterior paint on interior surface of closure	X	
	Surfaces not completely coated	X	
	Coating blistered, peeling, cracked, wrinkled or contains pin holes	X	
	Dirt or foreign matter imbedded in coating	X	
	Burned coating	X	
	Excess seam cement or flux not removed		X
	Dirt or foreign matter in can		X
	Welding flash not removed		X
	Melting splinters		X
Construction and workmanship (general)	Hole, crack, or fracture in metal	X	
	Component bent out of shape	X	
	Component missing	X	
	Vertical body seam assembled in front instead of at end opposite pouring spout		X
Closure	Cover not welded as specified	X	
	Cover not aligned with opening	X	
	Cover assembly excessively loose	X	
	Gasket wrong size or type	X	
	Gasket cement visible at gasket edges		X
	Locking lever cam sprung	X	
	Locking level cam fails to compress gasket (in closed position)	X	
	Hinge pin not peened or insecurely peened	X	
	Locking cam stud not peened or insecurely peened	X	
	3/16-inch radius of locking lever cam does not engage or seat with lock ring cam lugs	X	
Handle	Two or more spot welds missing	X	
	One spot weld missing		X

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Marking	Missing, incomplete, not legible, not specified type or size, not in proper location		X

4.4.4 End item dimensional examination. The end item shall be examined for conformance to the dimensions specified on Drawings 3-3-22 and 3-3-57, except the thickness of the walls of the steel or aluminum, as applicable, shall be measured in-process. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of water cans. The sample unit shall be one can. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 End item testing. The water can shall be tested for the characteristics listed in table V. The lot size shall be expressed in units of water cans. The sample unit for each test shall be one water can. The inspection level shall be S-2. Failure of any test shall be cause for rejection of the lot.

TABLE V. End item tests

Characteristic	Requirement paragraph	Test method	No. of determs. per sample unit	Results reported as
Handle spot welds	3.4.1.1 and 3.4.2	4.6.12	1	Pass or fail
Cover leakage	3.4.3	4.6.2	1	Pass or fail
Water taste change	3.5.3.2.2 and 3.5.3.4.1.1	4.6.3	1	Pass or fail
Impact flexibility	3.5.3.4.1.1	4.6.9	1	Pass or fail
Resistance to hot water	3.5.3.4.1.1	4.6.10	1	Pass or fail
Exterior coating thickness	3.5.3.4.2.1 and 3.5.3.4.2.2	4.6.111	1	Nearest 0.1 mil

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4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of bundles. The sample unit shall be one bundle fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components such as loose strapping, missing strapping, bulging or distortion of bundles.
Contents	Number of cans per bundle more or less than required.

4.4.7 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

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4.5 Certification compliance examination. Certifications, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the document requirements shall be examined and validated (see 3.3 and 3.3.6), as proof of compliance.

4.6 Methods of inspection.

4.6.1 Air pressure leakage test for types I and II. After complete assembly, all cans shall be tested for air leakage by being submerged in water for a sustained period of not less than 1 minute while filled with compressed air at a pressure of 8 pounds per square inch, to determine compliance with the non-leak requirements of 3.4.1.1 or 3.4.2, as applicable. Any non-conformance to specified requirements shall be cause for rejection of the can.

4.6.2 Cover leakage test for types I and II. The cans shall be filled with water and the cover assembly closed and locked. The can shall then be placed in an upside down position for not less than 30 minutes and examined for evidence of water leakage to determine compliance with the non-leaking requirements of 3.4.3. Any visible evidence of water leakage shall constitute failure of this test.

4.6.3 Taste test for types I and II. The cans shall be filled with distilled water or water containing less than 10 parts per million of impurities. The water shall be removed and the can left open and aired at room temperature for 24 hours. The can shall be filled with distilled water or water containing less than 10 parts per million of impurities at 120°F ± 5°F and maintained at this temperature for 24 hours with the cover closed but not locked. At the end of this time, the water shall be tasted and taste compared with water from the original sample to determine compliance with the neutral taste requirement of 3.5.3.2.2 or 3.5.3.4.1.1, as applicable. Any metallic, sulfurous, organic solvent or resin, or noticeable difference in taste between the water before testing and a sample of the same water after testing shall constitute failure of this test.

4.6.4 Resistance to acid. One test panel shall be immersed one-half its length in 0.1 percent normal sulfuric acid solution for 72 hours at 100°F ± 5°F. The panel shall be rinsed with clean, cold water and patted dry, and the coating on both sides shall be examined immediately to determine compliance with 3.3.6. The examination for softening shall be made by scraping with the thumbnail or a knifeblade in a continuous stroke through both immersed and non-immersed areas of the coating, and noting any difference between immersed and non-immersed portions. Any defects occurring in the coating within 0.25 inch from the edges of the panel shall be ignored.

4.6.5 Resistance to chlorine. The test shall be conducted as specified in 4.6.4 except that the test liquid shall be a solution of calcium hypochlorite containing 0.1 percent available chlorine at start of test.

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4.6.6 Resistance to tri-sodium phosphate. The test shall be conducted as specified in 4.6.4 except that the test liquid shall be a 1-percent solution of trisodium phosphate (calculated on the anhydrous basis) and the immersion time shall be 12 hours.

4.6.7 Resistance to steam. One test panel shall be exposed to the water vapor above the water surface in a covered heating vessel containing boiling water (e.g. steam kettle) at 212°F for 4 hours. The test panel shall be examined immediately upon cooling at room temperature, in the manner described in 4.6.4, to determine compliance with 3.3.6.

4.6.8 Resistance to freeze-thaw. One test panel shall be immersed horizontally midway in a depth of 2 inches of water which shall then be frozen solid. The ice shall then be melted by repeated applications of boiling hot water. The coating shall be examined in the manner described in 4.6.4 to determine compliance with 3.3.6.

4.6.9 Impact flexibility. The can to be tested shall be laid flat on a broad side and an 8-pound iron ball dropped on it from a height of 5 feet. The temperature of the can shall be 20°F +2°F. The same test shall be repeated on the other broad side with the temperature of the can between 70° and 90°F. During this test, the can shall be empty. The ball shall be dropped to land within the 8-inch square area occupied by the embossed X. The interior coating at the area of impact shall be examined for evidence of cracking, checking, or flaking to determine compliance with the requirements of 3.5.3.4.1.1. The examination shall be conducted by feeling with the pad of the finger and by scraping lightly with the fingernail and noting any difference in texture between the impacted area and adjacent area of coating. Any noncompliance with specified requirements shall constitute failure of this test.

4.6.10 Resistance to hot water. The can shall be filled with distilled or deionized water to a point not less than 0.125 inch above the welded shoulder seam and then placed on 2000-watt electric hot plate. The lid shall be closed, but to avoid possible steam explosion, shall not be locked. As soon as the water boils, the hot plate shall be turned off and the can allowed to stand and cool for 5 hours, with the lid remaining closed. The lid shall then be opened, and the can immediately emptied. The entire interior coating shall be examined for compliance with the nonsoftening, nonblistering, separation, loosening, rust, visible evidence of change, and penetration requirements of 3.5.3.4.1.1. A drum light and mirror shall be used to assist in viewing the inner surfaces. Examination for softening and loosening shall be conducted by scraping the coating with the thumbnail in comparison with the coating of an untested control can and noting any marked differences. Examination shall be made at random locations which, however, shall include the bottom, body, headspace, bottom seam, welded seams, neck, neck flange, neck seam, and underside of lid. General blistering within any single area in excess of 2 square inches, i.e. an area containing an accumulation of blisters of any size including pin-point, shall constitute failure of this test. General

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blistering of areas each less than 2 square inches but totaling more than 6 square inches shall constitute failure of this test. The lid and neck coating shall be examined first, and the examination of the entire can shall be completed within 5 minutes of emptying of the can, to preclude drying and disappearance of any blisters that may have formed. Any nonconformance with specified requirements shall constitute failure of this test.

4.6.11 Exterior coating thickness. A magnetic type gage shall be used for measuring exterior coating thickness requirements of the cans. Any nonconformance with specified requirements as specified in 3.5.3.4.2.1 or 3.5.3.4.2.2 as applicable shall constitute failure of this test.

4.6.12 Handle spot welds test for types I and II. An even pull of 500 pounds shall be applied to each handle bar. Evidence of any weld pulling away from the can shall constitute failure of this test.

5. PACKAGING.

5.1 Packing. Packing shall be level A.

5.1.1 Level A. Seven cans shall be strapped in a bundle as shown on Drawing 5-11-809.

5.2 Palletization. When specified, forty-two cans, packed as specified in 5.1, shall be palletized in accordance with load type III of MIL-STD-147. Six bundles of cans shall be positioned three bundles to a course and two courses high. The palletized load shall be provided with a fiberboard cap fabricated with grade V2s of PPP-F-320, with horizontal separator not required, and strapped by bonding means D, K, and L.

5.3 Marking. In addition to any special marking required by the contract, each bundle of water cans shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The cans are used for supplying water of sanitary quality in a sanitary manner to personnel of the Armed Services.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type of can required (see 1.2).
c. When a first article is required (see 3.1, 4.3 and 6.3).
d. When palletization is required (see 5.2).

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6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be in preproduction sample. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.2).

6.5 Subject term (key word) listing.

Can
Container
Storage tank
Tank

6.6 Changes from previous issue. Asterisks are not used in this revision to identify the changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - MC
Air Force - 99

Review activities:

Army - MD
DLA - GS

User activity:

Air Force - 84

Preparing activity:

Army - GL

Project No. 7240-0233

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-13984G	Cans, Water, Military, 5-Gallon

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP-1, Feb 86

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